Sunday Marathon: Probability From Basic

LIVE CLASS

Question of the Day: SBI PO MAINS

Direction (1-3) - There is three bags A, B and C. In each bag there are three types of coloured balls Yellow, Green and Black.

In bag A, no. of yellow-coloured balls are y and no. of green coloured balls are g. Number of green coloured balls are 4 more than the number of yellow-coloured balls. When one ball is picked at random then the probability of getting black colour ball is 5/13. The value of y is 18 2/11% less then g.

In bag B, number of yellow-coloured balls is 200/9% more than that of bag A. If two balls are picked at random from bag B then the probability of getting both green colour ball is 4/37. Total number of balls in bag B is 75.

In bag C, the ratio of number of green coloured balls and number of black coloured balls is 7 : 5. Total number of green and black coloured balls is 36. If one ball is picked at random then the probability of getting one yellow ball is 7/13.

Q1. If x number of yellow balls from bag B are taken and placed into bag A and 20% of black balls from bag A are taken and placed into in bag B. If we pick one ball from bag B then the probability that the ball is of black color is 11/26. Then find the value of x?

(a) 5 (b) 6 (c) 3 (d) 2 (e) None of these

Q2. If one ball picked at random from each of the bag A and bag B then find the probability that both of the balls are of the same color?

(a) 21×47/65×75 (b) 22×43/65×75
(c) 11×17/65×75 (d) Can't be determined (e) None of these

Q3. Difference between the number of green balls in bag A and bag C is how much percent more/less than the sum of the number of black balls in bag A and bag C together?

(a) 100% (b) 95% (c) 97.5% (d) 102.5% (e) None of these

Q. Find the total number of balls in Bag A and Bag C together? TITA

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Foundation

1. **In an examination, there are three sections namely Reasoning, Maths and English. Reasoning part contains 4 questions. There are 5 questions in maths section and 6 questions in English section. If three questions are selected randomly from the list of questions, then what is the probability that all of them are from maths?**
   A. 7/91
   B. 8/91
   C. 2/91
   D. 4/91

2 A basket contains 5 red 4 blue 3 green marbles. If three marbles picked up random, What is the probability that either all are green or all are red?
A. 1/20
B. 7/20
C. 3/20
D. 9/20

3.**A basket contains 5 red 4 blue 3 green marbles. If two marbles picked up random, What is the probability that both are red?**
A. 4/33
B. 5/33
C. 7/33
D. 8/33

4. A bag contains 5 red caps, 4 blue caps, 3 yellow caps and 2 green caps.If three caps are picked at random, what is the probability that two are red and one is green?
A. 22/55
B. 15/81
C. 10/91
D. 5/91

5.**A bag contains 5 red caps, 4 blue caps, 3 yellow caps and 2 green caps. If four caps are picked at random, what is the probability that two are red, one is blue and one is green?**
A. 22/1001
B. 80/1001
C. 21/1001
D. 55/1001

6. **Three Bananas and three oranges are kept in a box. If *two fruits are chosen at random*, Find the probability that one is Banana and another one is orange?**
   A. 1/5
   B. 3/5
   C. 4/5
   D. 2/5
   E. None of these

7.A basket contains 6 White 4 Black 2 Pink and 3 Green balls. If three balls picked up random, What is the probability that all three are White?
A. 4/91
B. 5/93

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C. 7/97
D. 8/92

7(B).**A basket contains 6 White 4 Black 2 Pink and 3 Green balls. If three balls are picked at random, what is the probability that two are Black and one is Green?**
A. 22/355
B. 15/381
C. 10/393
D. 14/455
E. 18/455

Q(8 –9) There are 6 players in a cricket which is to be sent to Australian tour. The total number of members is 12.

8. **If 2 particular member is always included**
   a) 210
   b) 270
   c) 310
   d) 420
   e) None of these
9. **If 3 particular player is always excluded**
   a) 76
   b) 82
   c) 84
   d) 88
   e) None of these

10.A basket contains 6 White 4 Black 2 Pink and 3 Green balls.If two balls are picked at random, what is the probability that either both are Pink or both are Green?
A. 2/105
B. 4/105
C. 8/137
D. 5/137
E. None of these

TYPE –ATLEAST)

Q1 Out of 14 applicants for a job, there are 6 women and 8 men. It is desired to select 2 persons for the job. The probabilty that atleast one of selected persons will be a Woman is?
A. 77/91
B. 54/91
C. 45/91
D. 40/91
E. None of these

2.**In a bag there are 4 white, 4 red and 2 green balls. Two balls are drawn at random.What is the probability that at least one ball is of red colour?**
A. 4/3
B. 7/3
C. 1/3
D. 2/3

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3.A basket contains 5 red 4 blue 3 green marbles. If three marbles picked up random, What is the probability that at least one is blue?
A. 41/55
B. 53/55
C. 47/55

4.**A basket contains 6 White 4 Black 2 Pink and 3 Green balls. If four balls are picked at random, what is the probability that atleast one is Black?**
A. 69/91
B. 80/91
C. 21/91
D. 55/91
E. None of these

5.A box contains 30 electric bulbs, out of which 8 are defective. Four bulbs are chosen at random from this box. Find the probability that at least one of them is defective?
a) 432/783
b) 574/783
c) 209/784
d) 334/784
e) None of these

New Type:

Question: 1

A box contains 27 marbles some are blue and others are green. If a marble is drawn at random from the box, the probability that it is blue is 1/3. Then how many numbers of green marbles in the box?
A. 10 B. 15 C. 14 D. 18

Question: 2

Bag A contains 7 Red Balls, 'X' Green Balls, and 5 Yellow Balls. The probability to pick Green Ball at random is 2/5. Another Bag B contains 'X-3' Red Balls, 'X-4' Yellow Balls and 6 Green Balls. If two balls are picked one after the other from Bag B at random then what is the probability for the Balls to be Red?
1. 1/21 2. 2/21 3. 3/21 4. 4/21 5. Cannot be determined.

Type: with replacement

Question1: A bag contains 6 red balls and 8 green balls. Two balls are drawn at random one after one with replacement. What is the probability that-

Q. Both the balls are green.
(a) 13/49
(b) 15/49
(c) 16/49
(d) 17/49
(e) None of these

Q. First one is green and second one is red.
(a) 16/49

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(b) 14/49
(c) 11/49
(d) 12/49
(e) None of these

Q. Both the balls are red.
(a) 14/49
(b) 9/49
(c) 11/49
(d) 12/49
(e) None of these

Type: without replacement

Question1

There are 4 black balls and 6 white balls. 2 balls are drawn one by one without replacement. What is the probability that the balls are same in colour?
(a) 9/21
(b) 8/17
(c) 5/14
(d) 7/15
(e) 9/19

Question2

A bag contains 5 red and 7 white balls. Four balls are drawn out one by one and not replaced. What is the probability that they are alternatively of different colours?
a) 7/99
b) 11/99
c) 14/99
d) 19/99
e) None of these

Type: at most

Question1: **There are 4 red balls, 5 white and 3 green balls in a basket. 3 balls are chosen at random. What is the probability that there is at most 1 green ball?**
(a) 13/40
(b) 48/55
(c) 25/68
(d) 8/33
(e) 9/19

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